

Probability Lecture 2

Robert O'Connell

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Conditional Probability

Use new information to revise a model

Defintion of Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Properties of Conditional Probabilities

- $P(A|B) \geq 0$
- $P(\Omega|B) = 1$
- $P(B|B) = 1$
- If $A \cap C = \emptyset$, then $P(A \cup C|B) = P(A|B) + P(C|B)$

We can see that conditional probablities satisfy all the standard probability axioms

The Multiplication Rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(A \cap B) = P(B)P(A|B) = P(A)P(B|A)$$

can be extended to N events

Total Probability Theorem

We have

- Partition of sample space into $A_1, A_2, \dots, A_n$
- Have $P(A_i)$ for every i
- Have $P(B|A_i)$ for every i

Then

$$P(B) = \sum_{i=1}^n P(A_i)P(B|A_i)$$

Bayes' Theorem

We have

- Partition of sample space into $A_1, A_2, \dots, A_n$
- Have $P(A_i)$ for every i
- Have $P(B|A_i)$ for every i

Then

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(A_i)P(B|A_i)}{\sum_j P(A_j)P(B|A_j)}$$

This is a systematic approach for incorporating new evidence

Bayesian Inference

Initial Beliefs $P(A_i)$ on possible cause of an observed event B

- Model if the world under each $A_i : P(B|A_i)$

$$A_i \rightarrow B \text{ using model \& } P(B|A_i)$$

- Draw conclusions about causes

$$B \rightarrow A_i \text{ using inference \& } P(A_i|B)$$

Bayesian inference reasons from observed effects to probable causes. Given initial beliefs $P(A_i)$ and forward models $P(B|A_i)$, we observe B and compute $P(A_i|B) \propto P(B|A_i)P(A_i)$. *Example:* observing fever (B), with causes $A_1 = \text{flu}$, $A_2 = \text{COVID}$, $A_3 = \text{cold}$, we combine prior probabilities $P(\text{flu}) = 0.3$, $P(\text{COVID}) = 0.1$, $P(\text{cold}) = 0.6$ with likelihoods $P(\text{fever}|\text{flu}) = 0.9$, $P(\text{fever}|\text{COVID}) = 0.8$, $P(\text{fever}|\text{cold}) = 0.3$ to determine which cause is most likely responsible.